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Land Use-Land Cover and Hydrological Modeling: A Review

Amirhossein Shadmehri Toosi | Okke Batelaan | Margaret Shanafield | Huade Guan

College of Science and Engineering, Flinders University, Adelaide, South Australia, Australia

Correspondence: Amirhossein Shadmehri Toosi (amirh.shadmehr@flinders.edu.au)

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ABSTRACT

Given the extent of recent land use-land cover changes and the consequences for hydrological effects, it is important to capture the role that land use-land cover plays in hydrological model conceptualization. This need is amplified under a changing climate, as the hydrological sensitivity of catchments to climate change can vary depending on the land use-land cover. Traditionally, model calibration has prioritized parameter adjustment rather than enhancing the representation of land use-land cover processes. This practice—though useful in improving model fit—can lead to misinterpretations, especially when simulating under varying land use-land cover and climate conditions. While airborne and satellite remote sensing technologies now allow for frequent, large-scale monitoring of land surface parameters, their spatiotemporal complexities remain underutilized in hydrological models. By analyzing the conceptualizations of land use-land cover related hydrological processes within models, this paper reviews the progress in the current state of modeling approaches and discusses future research needs and directions. The path ahead for land use-land cover and hydrological modeling is one of embracing technological advancements and methodological rigor, focusing on representing the multifaceted reality.

1 | Introduction

The extent and nature of different land cover types and how people utilize the landscape are documented by land use and land cover (LULC) data. Land cover refers to the physical materials on the earth's surface, such as vegetation, water bodies, and artificial structures, which can be identified through satellite and aerial imagery analysis (Green et al. 1994). On the other hand, land use denotes how these land covers are utilized by humans, a less directly observable dimension from remote sensing techniques (Green et al. 1994). Despite their differences, land use and land cover are often analyzed as LULC changes, reflecting their intertwined roles in shaping environmental outcomes. LULC change has significant implications for hydrological processes, among other physical, ecological, and socioeconomic dimensions (Li, Si, et al. 2018; Teklay et al. 2019; Zeng et al. 2020). For example, constructing a dam can have important local to regional impacts on water balances, or deforestation can impact carbon cycling (Dile et al. 2018; Li, Piao, et al. 2018). Understanding the effects of LULC change on the hydrological processes is needed to optimize natural resources management (Scanlon et al. 2005). Documenting the spatial patterns and determining the magnitude of LULC change effects is essential for understanding and managing hydrological systems (Cheng and Yu 2019). Moreover, the global impact of LULC change on hydrological processes may surpass the impact of climate change or could be amplified by it (Aghsaei et al. 2020). Therefore, a better

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understanding of LULC's role in hydrological processes is vital to employing adaptive measures to minimize the inevitable adverse effects of change in LULC (Cartwright et al. 2017; Sun et al. 2017).

Many different types of hydrological models that incorporate LULC have been used to simulate the implications of LULC change (Dwarakish and Ganasri 2015; Dey and Mishra 2017). The four primary processes affected by LULC are radiation energy partitioning, water interception dynamics, on-ground water partitioning, and subsurface processes. Hydrological models represent these processes with different levels of detail and complexity and allow simulations at various spatial and temporal scales. Although a simplified integration of LULC provides valuable insights into the functioning of hydrological systems, the underlying assumptions and simplifications used for simulating hydrological processes may introduce inaccuracies in the results (Beven 2018, 2019a). Yet, there are challenges to realistically incorporate each of these processes.

Traditionally, most hydrological models represent a highly simplified version of the interaction between vegetation and water. Despite major advances in our understanding of hydrologically relevant vegetation processes and properties in recent years, we lack integration of these features in hydrological models (Sun et al. 2017). Furthermore, even with the advancements in airborne and satellite remote sensing techniques for monitoring land surface parameters, the temporal dynamics of LULC have not been fully integrated into models. While current data availability permits LULC variations over time, models often assume constant land use for extended periods, spanning months or even years. To accurately capture the interactions between LULC, climate, and hydrology, tighter temporal integration of LULC dynamics and hydrological processes is required (Castillo et al. 2014). Although there are many reviews of the state of the art of hydrological modeling, an overall picture of the current status of LULC characterization in the processes used in hydrological models does not exist (Sood and Smakhtin 2015; Noszczyk 2019; Moges et al. 2020; van Stan and Friesen 2020).

Therefore, in this work, we start with four key processes influenced by LULC and a brief overview of the physical mechanisms affected by LULC. Further, we review how these processes are typically represented within hydrological models and discuss the associated challenges and limitations. With this foundation established, we guide the reader through a discussion on the current LULC data sources and their limitations. We emphasize the importance of selecting an appropriate model to investigate LULC impacts and enhancing the LULC representation within the model process to better align with study objectives. Finally, we delve into current model evaluation and calibration techniques and discuss ways to refine them. Readers are encouraged to refer to the Supporting Information for the overview of the main hydrological model types and their respective strengths and limitations (Table S1). We believe this work helps better evaluate the current gaps in hydrological modeling practice and the needed future developments, and it promotes best practices in modeling LULC impacts.

2 | Review of Main Hydrological Processes Affected by LULC

2.1 | Radiation Energy Partitioning and Surface Energy Balance

2.1.1 | Physical Mechanisms

LULC influences solar radiation partitioning through alterations in surface albedo, landscape thermal properties, and vegetation cover dynamics. This impact extends to sensible and latent heat fluxes, plant growth, wind patterns, and snowmelt (Zhang et al. 2010). Variations in vegetation can shift surface radiative properties and contribute to greenhouse gas emissions, notably from activities such as deforestation. Consequently, energy partitioning is influenced by local conditions including vegetation type and density. Specific ecosystem features such as soil type, water availability, and biodiversity, along with daylight hours and altitude, further influence this energy partitioning (Hammerle et al. 2008; Zhang et al. 2010). Such shifts can impact local climate and surface energy balance, underscoring the significant influence of LULC on radiation balance (Duveiller et al. 2018).

Figure 1 illustrates the energy balance in a terrestrial ecosystem, driven by solar radiation and various vegetation and soil energy fluxes. Radiation splits into direct (solar) and diffuse (reflection and emittance) forms. Atmospheric transmissivity, governed by cloud cover and pollution, dictates the fraction of solar radiation reaching the Earth. Solar radiation reaching the terrestrial system, shown by a large orange arrow, is the primary energy source, partially reflected back into the atmosphere as diffuse radiation (yellow) while the remainder is absorbed. This absorbed energy contributes to several processes, including sensible heat flux (red), which transfers heat from the surface

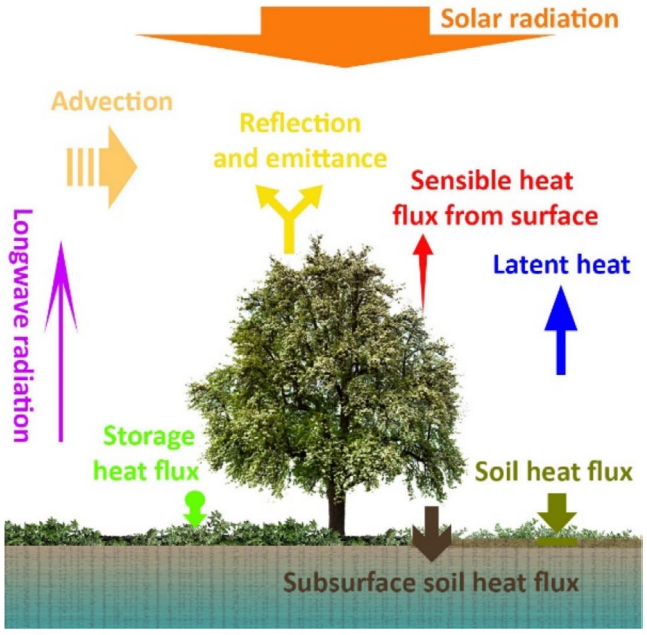


FIGURE 1 | Schematic picture of surface energy partitioning by vegetation.

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to the air, directly affecting local air temperature. Latent heat flux (blue) represents energy used for water evaporation and transpiration, integral to the hydrological cycle. Longwave radiation (purple) is emitted as thermal energy from the Earth's surface, cooling it by releasing energy into the atmosphere. Soil heat flux (olive green) moves energy from the surface into the upper soil layers, influencing soil temperature, while subsurface soil heat flux (dark brown) distributes heat within deeper soil layers. Advection (mustard yellow) introduces external heat and moisture, modifying local conditions. Finally, storage heat flux (green) represents the energy retained within vegetation and soil, moderating temperature fluctuations and supporting ecosystem processes by absorbing and releasing heat as needed. The discrepancy between incoming and outgoing radiation, termed net radiation, is vital for evaporation, snowmelt, and soil heat flux.

2.1.2 | Modeling Approaches

The global water budget is intrinsically linked to the global energy budget due to its impact on processes like evapotranspiration, snowmelt, and soil heat dynamics (Wang et al. 2018). When water changes phase (solid, liquid, and gas), energy is absorbed or released, thus affecting the energy budget (Healy et al. 2007). Incoming solar radiation is the fundamental energy driving the processes of photosynthesis, evaporation, heating of the soil, and energy storage in vegetation (Gu et al. 2005). Evaporation is the phenomenon by which water is converted from its liquid into its vapor phase, regardless of whether it resides in the soil, on the surface (including plants) or in vegetation. A simple energy budget for the Earth is (Sellers 1969):

$$R_n = G + \mathrm{LE} + H \tag{1}$$

where R_n is net radiation, the sum of incoming solar and longwave radiation minus reflected solar and emitted longwave radiation; G is the surface heat flux, that is, the energy used to warm soil or water in the case of a surface-water body; LE is the latent heat flux, that is, the energy used to evaporate water; and H is the sensible heat flux, or the energy used to warm air (Healy et al. 2007). Equation (1) states that available energy at the Earth's surface goes to heating the surface, warming the air, and evaporating water (Figure 1). Latent heat flux is the product of latent heat of vaporization (λ) and evapotranspiration rate (ET); that is, $\mathrm{LE} = \lambda \mathrm{ET}$. Evapotranspiration directly links the energy and the water budget because it appears in both.

In hydrological models, radiation energy partitioning and surface energy balance are pivotal in shaping processes such as evapotranspiration, soil moisture dynamics, snowmelt, and vegetation growth. These processes, in turn, have a significant impact on water temperature and surface runoff. Thus, vegetation dynamics inevitably affect or provide feedback to surface energy partitioning and the water budget (Forzieri et al. 2020; Lan et al. 2021). The following subsections will focus on evapotranspiration and snow processes, as these are critical elements of energy partitioning.

2.1.2.1 | Evapotranspiration. Evapotranspiration, in conditions without snow, refers to the collective water vapor from direct evaporation from the soil's surface layer, transpiration from both the roots and canopy of vegetation, and the evaporation of precipitation captured by the plant canopy (Singh 2017). Hydrological models typically handle evapotranspiration by either accepting it as input or calculating it as an output parameter. These models may treat evaporation in several ways: as a constant, as a set of monthly average values, as a user-defined time series of values on different temporal scales (monthly/seasonal), or compute it based on climatological variables like temperatures, wind, and sunshine hours (Bormann et al. 1996; Li et al. 2009; Zhang et al. 2016; Singh 2017; Chia et al. 2020).

There are two groups of evapotranspiration estimation methods in hydrological models: one first estimates separately open water evaporation, soil evaporation, and vegetation transpiration and then integrates them to calculate actual evapotranspiration. The other one first estimates potential evapotranspiration and then converts it into actual evapotranspiration. Zhao et al. (2013) called the first classification gathering methods and the second integrated converting methods. Integrated converting methods are typically applied in lumped conceptual and distributed models. Conversely, physically based hydrological models often employ classification gathering methods for estimating basin evapotranspiration; for more details, refer to Zhao et al. (2013). However, it is crucial to acknowledge that these classifications are not exhaustive, and various models may employ alternative methods or a combination thereof.

The ratio of the actual over the potential evapotranspiration changes with water availability in the soil. Hydrological models commonly utilize soil moisture extraction functions to depict the relationship between actual and potential evapotranspiration. Based on the function employed, it can be associated with the soil type and the Leaf Area Index (LAI). The LAI is intricately linked to the growth stage of the vegetation (Ritchie 1972; Mintz and Walker 1993).

The potential evapotranspiration estimation methods can be divided into energy-based, temperature-based, and mass transfer-based methods, depending on their mechanisms. Consequently, they vary strongly in data needs. Widely used methods are Hargreaves and Samani (1985), Priestly and Taylor (1972), and Monteith (1965).

The Penman-Monteith method requires air temperature, relative humidity, solar radiation, and wind speed. The Priestley-Taylor method requires solar radiation, air temperature, and relative humidity. The Hargreaves method requires air temperature only. Independent "ground truth" data is preferably used as a benchmark to assess the performance of these models. Eddy Covariance (EC) systems are widely regarded as the premier method for continuous evaporation measurement (Wang and Dickinson 2012) and are used worldwide, for example, FLUXNET (Baldocchi et al. 2001).

2.1.2.2 | Snow Process. The most common snow modeling approaches are based on either conceptual (temperature-based models) or physically based models (energy balance-based models) (Singh 2017; Ren and Liu 2019). Bucket-type snow models are typically based on a temperature-index approach, assuming a direct relationship between temperature and snow accumulation

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and melt. Most of these models aggregate data spatially into areas with similar hydrological responses, known as Hydrological Response Units (HRUs), which facilitate faster computational times and reduce the amount of input data required. HRUs are defined through a detailed hydrological systems analysis using Geographic Information System (GIS) techniques (Flügel 1995). This analysis incorporates key physiographic basin properties, including topography, soils, geology, rainfall patterns, and land use (Freudiger et al. 2017). By integrating these diverse factors, GIS helps ensure that each HRU accurately reflects the unique hydrological characteristics of its area, facilitating more precise and effective hydrological modeling.

Extended temperature-index approaches that include spatially distributed solar radiation are frequently used in hybrid methods, reducing the computational effort (Hock 1999). The inclusion of solar radiation is particularly relevant for high elevations, often glacierised areas, where temperatures seldom rise above freezing. This adaptation enhances model accuracy by accounting for the significant role of solar radiation in melting processes under consistently cold conditions (Pellicciotti et al. 2014; Gabbi et al. 2014).

More complex models are evolving from uniform snow cover models to more complex ones, allowing non-uniform cover due to topography, drifting, shading, and land cover. These models are generally highly complex, distributed, physically based, and very detailed, with a high spatial and temporal resolution. All energy fluxes are considered in these models (Lundberg and Halldin 2001). Still, their use is thus constrained by data availability and computational time, rendering them most suitable for small catchments or plot scales and limited durations, such as a few winter seasons or specific events only (Lundberg and Halldin 2001).

2.1.3 | Challenges

Differences between evaporation models and observations stem from both conceptual limitations in the models and flaws in measurement techniques (Coenders-Gerrits et al. 2020). In his 2004 article, Savenije criticizes the term "evapotranspiration" deeming it outdated due to its oversimplified grouping of diverse evaporative processes such as evaporation from intercepted water, transpiration from plants, soil evaporation, and open-water evaporation. He notes that these processes vary significantly in their temporal dynamics, physical properties, climatic interactions, and isotopic compositions, advocating for a more nuanced approach in hydrological studies (Savenije 2004).

In this context, the complexity of accurately modeling evaporation becomes evident when considering the role of atmospheric composition in influencing radiative fluxes, as highlighted by Dhara (2020). These fluxes significantly affect the earth's surface heating, which impacts evaporation rates. Similarly, Yin et al. (2019) emphasize the critical role of evaporative cooling on the earth's surface, which modulates radiation conversion into sensible heat, affecting temperature readings and further complicating evaporation modeling. These insights emphasize the complex interplay between atmospheric conditions and evaporation, reinforcing the need for sophisticated modeling approaches that can accurately reflect these dynamics.

In most existing studies, water budget and surface energy partitioning were generally examined in isolation regarding vegetation greening (Lan et al. 2021). However, these elements should not be analyzed in isolation due to their close interdependencies with LAI-related biophysical processes (Pitman 2003; Bagley et al. 2017; Zeng et al. 2017). The continuous rise in LAI, a result of vegetation greening, impacts surface albedo, thereby influencing the net terrestrial radiation (Rn) that is crucial for driving both latent and sensible heat fluxes (Kala et al. 2014). This, in turn, modifies the surface energy partitioning (Kleidon 2019; Forzieri et al. 2020). Therefore, a detailed examination of how surface energy partitioning and water budgets respond to vegetation dynamics amid widespread greening is essential (Kala et al. 2014; Ma et al. 2017).

Though the Penman-Monteith method is the standard for modeling potential evapotranspiration, its detailed process might not align with other models' accuracies, especially concerning soil moisture extraction functions (Neitsch et al. 2011). Some soil moisture extraction methods consider factors like root suction and canopy density, yet they often fail to account for spatial and temporal variations in surface characteristics. This oversight can lead to significant errors when calculating average daily soil moisture values (Baier and Robertson 1966). Moreover, using the Penman-Monteith equation with mean daily climatic parameters can result in substantial inaccuracies. Such errors are due to diurnal variations in wind speed, humidity, and net radiation, which are conditions not captured by daily averages, leading to potential discrepancies in evapotranspiration estimates (Neitsch et al. 2011). Hence, we need further research on how to select appropriate potential evapotranspiration equations and soil moisture extraction functions for different hydrological models to reduce their uncertainty.

Operational hydrology simplifies snow interception, even though snow storage, both in mass and duration, typically exceeds that for rain, making it a critical model component. The evaporation of snow intercepted by forest canopies remains a poorly understood element of the winter water balance equation (Lundberg and Halldin 2001). Ignoring it in hydrological models can alter the timing and volume of simulated snowmelt and affect the catchment water balance (Freudiger et al. 2017). In catchment-scale hydrological modeling, simulating separately the snow cover accumulation inside and beneath the canopy has rarely been applied so far, so more effort is needed in modeling snow processes (Gouttevin et al. 2015; Förster et al. 2018).

The effects of the forest canopy on snow accumulation and snowmelt processes need to be considered in simulations of the hydrological response of catchments with a significant fraction of the forested area. Despite the well-established accuracy of process-based, energy-budget snowmelt models, there is a propensity towards using temperature-index or degree-day snowmelt relationships in hydrological models to simulate snowpack processes (Walter et al. 2005). Yet, the critical temperature of these methods is not constant across basins; it varies with topography and elevation (Walter et al. 2005; Ren and Liu 2019). Models must also consider variations in canopy albedo due to its significant impact on snow interception (Lundberg and Halldin 2001). Oddly, many models employ energy balance approaches, like the Penman equation, for estimating potential

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evapotranspiration. Thus, it is inconsistent that these models do not employ the same energy balance approaches for modeling snowmelt (Walter et al. 2005).

The dynamics of precipitation characteristics, including type, duration, spatial distribution, and intensity, can significantly impact evaporation. Additionally, the interaction between precipitation and interception effects on sublimation and condensation poses challenges for accurate modeling (Svoma 2016; Gutmann 2020). Sublimation from canopies in winter, although less in volume than summer evaporation, plays a more significant role than evaporation in modeling winter hydrological processes, and measurements are scarce (Frank et al. 2019). Energy budgets have challenges explaining snow interception losses, as they need to account for melting, evaporation, and sublimation (Molotch et al. 2007; Allen et al. 2020). Incorrect snowmelt modeling affects not only the day on which the melt event was predicted inaccurately but also the day of the actual melt event, resulting in multiple periods of incorrect streamflow simulation (Zeinivand and Smedt 2009). Hence, the accurate spatial distribution of snow depth is vital for precise runoff calculations (Ren and Liu 2019).

2.2 | Water Interception Dynamics

2.2.1 | Physical Mechanisms

Interception refers to the portion of rainfall captured by the Earth's surface (e.g., vegetation). Part of this intercepted water evaporates back into the atmosphere (Muzylo et al. 2009; Meili et al. 2021). LULC, rainfall patterns, and evaporative demand influence the rate, capacity, and loss of interception (van Meter et al. 2016; Zhou 2022). While built-up areas experience interception, its significance is more pronounced in vegetated regions. With their vast surface area, trees, shrubs, and litter capture a large volume of rainwater, shielding the soil from direct impact (Figure 2). This conserves soil integrity and curtails erosion while modulating water flow to streams, thereby reducing flooding potential (Gerrits and Savenije 2011).

Canopy water storage varies based on intrinsic factors like vegetation density and plant morphology, and extrinsic factors like storm conditions (intensity and duration) and temperature (van Stan and Friesen 2020). These factors differ across plant species, leading to diverse canopy storage capacities (Klamerus-Iwan et al. 2020). Similarly, these factors influence a canopy's ability to store snow. Snow storage in canopies surpasses rain and has been observed to cover a significantly broader spectrum and seasonal variation depending on storm and climate conditions (Lundberg and Halldin 2001; Storck et al. 2002; Förster et al. 2018). Several extrinsic factors impact the response and capacity of the canopy reservoir, like hydrometeorological (storm) conditions that control crystal form, internal bonding of snow, and adhesion of snow to vegetation (Satterlund and Haupt 1967; Schmidt and Pomeroy 1990; Klamerus-Iwan et al. 2020). As a large proportion of intercepted water evaporates from the canopy, the understorey and litter layer can also significantly affect subsequent hydrological processes (Gerrits and Savenije 2011; van der Ent et al. 2014; Porada et al. 2018).

The ground layer's characteristics vary by region and ecology, ranging from short vegetation to decomposing organic materials or bare soil (Gerrits and Savenije 2011). These variations significantly influence local water processes, including precipitation interception and evaporation (Gerrits and Savenije 2011).

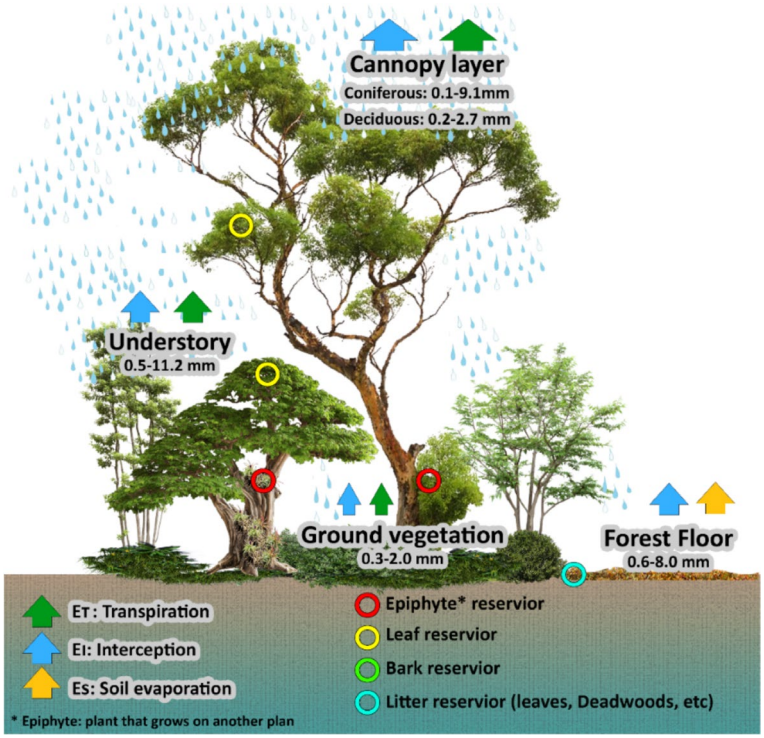


FIGURE 2 | Schematic representation of the multi-layer reservoirs within vegetated ecosystems, highlighting the water storage and movement dynamics (Reservoir capacities are based on Klamerus-Iwan et al. (2020)).

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The litter layer, often lacking deep-rooted vegetation, primarily experiences water movement through gravity and evaporation, with minimal transpiration (Klamerus-Iwan et al. 2020). This layer serves as a transitional zone, impacting water storage through variations in litter composition and its seasonal dynamics (Gerrits and Savenije 2011; van Stan et al. 2017; Klamerus-Iwan et al. 2020). Hence, conceptualizing the water storage at the land surface is challenging due to the litter layer, but also due to the variability in precipitation type and intensity (Gerrits et al. 2007, 2010; Klamerus-Iwan et al. 2020). Although forest floor evaporation seems to overlap with soil evaporation, the distinction is that soil evaporation pertains to the water in the root zone (de Groen and Savenije 2006).

In forest ecosystems, microclimates significantly influence hydrological processes. Wind plays a role in understory evaporation but has a limited impact on litter storage due to vegetation's wind-reducing effects (Singh 2017). The forest floor experiences less evaporation than the overstory due to lower radiation, yet intermittent storms can heighten evaporation, affecting litter storage and throughfall (Gerrits et al. 2009; Gerrits and Savenije 2011). Additionally, snow dynamics under the canopy are complex. Canopies can alter ground snow processes, causing non-uniform snow patterns and influencing melt timing (Förster et al. 2018). Snow interception by dense canopies and the subsequent sublimation or melt of intercepted snow can reduce sub-canopy snow accumulation by up to 60% (Hardy et al. 1997). Factors like shading and reduced wind contribute to delayed melting, while increased tree-emitted radiation accelerates it (Sicart et al. 2004; Essery et al. 2008; Pomeroy et al. 2009; Strasser et al. 2011; Lundquist et al. 2013).

The primary factors influencing snow processes are canopy density and specific geographical attributes such as elevation, slope, and weather conditions (Pomeroy et al. 2002; Jost et al. 2007; Strasser et al. 2011). In contrast to the influence of phenology and morphology on the movement of liquid water, the discharge of intercepted snow is predominantly determined by hydrometeorological factors rather than the specific characteristics of individual trees. However, these relationships have not been extensively researched (Satterlund and Haupt 1970; Schmidt and Gluns 1991; Gouttevin et al. 2015). These nuanced interactions underscore the complex interdependencies within forest hydrology, shaped by the delicate balance of microclimatic factors and vegetative structures.

2.2.2 | Modeling Approaches

In large-scale modeling, the concept of interception predominantly pertains to vegetated areas. Considering that the proportion of urban areas is typically minimal in comparison to other LULC categories, it is widely accepted that, over extended periods like a year, rainwater intercepted by ground surfaces and building roofs tends to evaporate fully (Daba and You 2020; Zhou 2022). Therefore, the runoff coefficient determines solely the rainwater interception evaporation on sealed ground and pavement (Zhou 2022).

Numerous models have been developed for simulating interception, especially in forested contexts. These models can be adapted to account for interception on the forest floor or any other surface (Gerrits and Savenije 2011). The modeling of this interception incorporates diverse approaches, each characterized by its unique methodology (Muzylo et al. 2009). The Gash Analytical Model (Gash 1979) is a prominent example among models frequently utilized in hydrological studies, known for its grounding in observed data and statistical correlations. This model, and others like it, employs parameters such as total precipitation, canopy evaporation, throughfall, and storage capacity. While the strength of the Gash Analytical Model lies in its simplicity and robust empirical basis, this simplicity may also limit its comprehensive applicability in capturing all aspects of hydrological processes (Gash 1979). It is important to note that this limitation is not unique to the Gash Analytical Model but is a characteristic shared by similar models in the field.

Conceptual models usually deal with redistributing rainfall volume using a mass balance equation. For example, the Rutter model uses a bucket-type approach to divide rainfall into intercepted and throughfall components based on canopy storage capacity (Gash and Morton 1978). Stochastic models in this category may use Markov Chains to represent the probabilistic nature of rainfall (de Groen 2002). Conceptual models provide a more structured representation compared to empirical models, dividing the canopy into different layers and using parameters like total precipitation, storage capacity, and evaporation rate to estimate interception, storage, and evaporation processes (Muzylo et al. 2009).

Physical models are the most detailed, often employing differential equations to simulate water flow over leaves, infiltration, and evaporation (Muzylo et al. 2009; Paniconi and Putti 2015). These models may be three-dimensional and incorporate variables such as wind speed, temperature, relative humidity, LAI, and storage capacity. These parameters are crucial for accurately modeling the complex behaviors involved in water interception by vegetation, including how water is captured, stored, and eventually evaporated or absorbed by the plant (Herwitz and Slye 1995; Bussière et al. 2002; Muzylo et al. 2009).

2.2.3 | Challenges

Operational hydrology often oversimplifies the interception process, potentially leading to evaporation underestimation in hydrological models (Lindström et al. 1994; Lundberg and Halldin 2001; Mueller et al. 2013; Liu et al. 2016). Modeling interception is challenging due to the complexities of landscapes. While short vegetation is frequently reasonably modeled, forests are multi-layered and have varying water interception rates across layers, which present modeling challenges (Breuer et al. 2003; Gerrits and Savenije 2011). Comparing different forest evapotranspiration models with Eddy Covariance (EC) data has shown the highest discrepancy, indicating that forest systems are likely not yet modeled correctly (Ershadi et al. 2014; Ha et al. 2015).

In vegetated areas, the mechanisms of water interception are often modeled and simplified as the dynamic processes of filling and emptying static water storage compartments. Most models conceptualize the storage and redistribution of

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precipitation water on vegetation as a uniform-sized reservoir, or, more recently, the LAI is used by models to compute the maximum storage at any time in the land cover/crop growth cycle. However, accurately measuring and quantifying LAI is challenging due to sensor spectral and spatial resolution limitations and the variability in vegetation structure and density (Luo et al. 2020; Xu et al. 2020; Chen, Li, et al. 2023; Li et al. 2023).

Numerous hydrological models directly correlate canopy storage with the LAI, adjusted by a daily phenological multiplier specific to different plant functional types (Gerten et al. 2004; Murray 2014; Klamerus-Iwan et al. 2020). The storage capacities indicated by these models are generally low (0.2–2 mm) compared to field measurements (1–16 mm), likely due to the limitation of LAI in encompassing the diverse other storage components present within vegetated ecosystems (Gerrits and Savenije 2011; van Stan and Pypker 2015; Porada et al. 2018; Klamerus-Iwan et al. 2020). Vegetated ecosystems possess interconnected reservoirs with different capacities and operational timescales. Components such as tree crowns, trunks, understory, and litter are often overlooked in models, although their storage capacities can be significant (Klamerus-Iwan et al. 2020).

In multilayer systems (e.g., forested systems), multiple storages' filling and spilling characteristics induce temporal shifts (Gerrits et al. 2010). Moreover, the potential evaporation below the canopy is typically lower than that above the canopy. Coupled with the generally larger storage capacity of the forest floor, this leads to residence times for forest floor interception ranging from several hours to days. In contrast, water intercepted by the canopy typically has a residence time of less than an hour (Levia et al. 2011; Wang-Erlandsson et al. 2014; Li et al. 2017; Coenders-Gerrits et al. 2020).

In fact, modeling the sequence in the understorey and forest floor storage and the uncertainty in potential evaporation are the major challenges in modeling the forested setting (Coenders-Gerrits et al. 2020). In addition, there is high seasonal variation and complexity in vegetated systems. For example, using LAI methods, the wintertime interception is close to zero for deciduous trees; however, the forest floor can still have significant water content that can be evaporated (Gerrits et al. 2010; Gerrits and Savenije 2011; van Stan et al. 2017). Furthermore, if the growing season's length is not considered in the model, apparently, the vegetation transpires more efficiently than would be expected from air temperature alone (Blöschl and Montanari 2010). The amount of actual evapotranspiration contributed by intercepted rainfall can be significant, especially in forests where, in some instances, evaporation of intercepted rainfall is greater than transpiration (Neitsch et al. 2011).

It is commonly assumed that interception in leafless deciduous vegetation is low due to rapid unloading under dry snow conditions. However, this may contrast with conifers, where denser intercepted snow can lead to significant snow storage. This discrepancy can result in models overestimating the amount of water entering the unsaturated zone. Consequently, transpiration rates may be inaccurately high, either in reality or as a result of calibration (van den Hoof et al. 2013). However, this suggests that the model is fundamentally flawed, carrying significant implications for research involving, for instance, climate and/or change modeling (Coenders-Gerrits et al. 2020). To accurately depict snow processes in forested areas, it is essential to adopt a comprehensive approach that simulates both the snow melt rate and the timing of meltwater release (Whitaker and Sugiyama 2005; Zheng et al. 2016; Förster et al. 2018). The potential benefits of incorporating intercepted snow evaporation into models have been highlighted in recent studies, suggesting a possible route to refining their accuracy (Lundberg and Halldin 2001; Goel 2011; Xiao et al. 2019; Bouchard et al. 2024).

2.3 | Water Partitioning Between Runoff and Infiltration

2.3.1 | Physical Mechanisms

Upon reaching the ground, precipitation partitions into runoff and infiltration. Both processes are influenced by LULC following precipitation or melting events. Vegetation and litter layers protect soil from raindrop splashes by intercepting precipitation, preventing surface sealing and crusting of soil, extending the time of soil infiltration, and enhancing sediment deposition by increasing soil surface roughness (Li et al. 2014). Additionally, factors like terrain, slope, geology, rainfall characteristics, and climate play crucial roles in determining runoff and infiltration, with variations often linked to LULC (Bhark and Small 2003; Wu et al. 2021).

Human activities associated with different LULC patterns significantly influence on-ground water partitioning. For instance, agricultural practices such as tillage can modify infiltration patterns, either retaining or directing water (Hangen et al. 2002). In urban areas, impermeable surfaces like roads increase runoff and reduce infiltration (Shuster et al. 2005). Urban storm sewers can also amplify streamflow, potentially leading to flooding (Shadmehri Toosi et al. 2019). Infrastructure developments, such as dams, play a role in altering outflow and the volume of water downstream (Schmutz and Moog 2018).

Moreover, changes in LULC, such as deforestation or the expansion of agriculture, expose soil to erosion. This erodes the nutrient-rich topsoil and diminishes its infiltration and retention capacities (Belay and Mengistu 2021; Alem 2022; Prashanth et al. 2023). As erosion continues, it degrades the soil, forming surface crusts that further impede infiltration. The eroded particles accumulate in depressions and streams, elevating runoff levels. This process can lead to the formation of gullies or rills, which act as fast channels for water, reducing the time available for infiltration. Without the protective cover of vegetation, the health of soil biota, essential for maintaining soil quality, deteriorates. Overall, these LULC-induced alterations result in increased runoff. Reforestation, on the other hand, typically increases evapotranspiration, which can reduce groundwater recharge and streamflow. However, it may also enhance soil infiltration in certain conditions, potentially increasing streamflow during dry periods and altering its seasonal distribution. Predicting the impacts of reforestation on streamflow is complex and depends on site-specific factors such as climate, soil properties, forest characteristics, and underlying hydrogeology (van Meerveld and Seibert 2024).

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2.3.2 | Modeling Approaches

Hydrological models employ diverse techniques to partition rainfall into runoff and infiltration (Sitterson et al. 2018). Empirical models are best suited for ungauged watersheds where runoff is the primary output of interest due to their simplicity, quick computational turnaround, and cost efficiency (Dawson and Wilby 2001; Pechlivanidis et al. 2011). A typical example is the Curve Number (CN) method, developed by the USDA (Mishra and Singh 2003). It provides a runoff estimate predicated on soil attributes and overarching LULC (Kokkonen et al. 2001; Devia et al. 2015; Sitterson et al. 2018). Another example is the Rational Method, which is used to estimate peak flow rates based on the runoff coefficient, rainfall intensity, and drainage area. While the Rational Method incorporates a runoff coefficient, it does not have the same level of detail or methodology as the CN method (Rossi 1994). Models, such as Horton's equation designed for infiltration assessment, indirectly aid in runoff estimation. In its original form, Horton's equation does not explicitly include LULC like the CN method. However, the parameters of Horton's equation can be adjusted based on LULC and other factors in practical modeling applications (Horton 1941). The limitations of empirical models include their dependence on field observations, which are sometimes unavailable (Morbidelli et al. 2018).

Conceptual models like HSPF, TOPMODEL, HBV, Stanford, and ABCD are on the other end of the spectrum, offering a simplified representation of physical hydrological processes (Crawford and Linsley 1966; Bergstrom 1976; Beven and Kirkby 1979; Thomas 1981; Bicknell et al. 1997). Conceptual models view a catchment as interconnected storage units with flow movements defined by mathematical functions (Jehanzaib et al. 2022). A significant limitation is that their parameters are not directly derived from catchments, necessitating calibration (Madsen 2000). In the HSPF model, different LULC types are represented through land segments, each with its distinct set of parameters tailored to the hydrological response of that specific LULC (Bicknell et al. 1997). TOPMODEL integrates LULC by influencing the topographic wetness index with different land covers, affecting parameters like evapotranspiration rates (Beven et al. 2021). The HBV model adopts a distributed approach, where different LULC types are represented by unique response units, each characterized by its hydrological parameters (Bergstrom and Forsman 1973). The Stanford Watershed Model employs land use coefficients to adjust hydrological parameters in accordance with the specific LULC type (Crawford and Linsley 1966). Lastly, the ABCD hydrological model simulates the rainfall-runoff process in a catchment using four parameters: A, B, C, and D (Thomas 1981). The model can account for the effects of LULC changes on hydrology by adjusting the values of these parameters according to the characteristics of different LULC types.

Physical (process-based) models employ differential equations to simulate catchment behavior over space and time, as highlighted in the review by Devia et al. (2015). These models intricately describe mass and momentum balances for individual grid cells or sub-catchments and incorporate boundary conditions to account for interconnectivity (Todini 1988). A key aspect of these models is their approach to modeling infiltration and runoff processes. Infiltration is typically modeled by considering the soil's capacity to absorb water, which is influenced by soil texture, structure, and moisture content. This can involve the use of equations like the Richards equation to simulate the movement of moisture through unsaturated soils. On the other hand, runoff is modeled based on the excess water that cannot infiltrate, often using approaches like the kinematic wave model or the diffusion wave model to represent overland flow dynamics (Orlandini and Rosso 1996; Singh 1997; Kim et al. 2012). While these models do not require extensive meteorological and hydrological data for calibration, they do necessitate a detailed assessment of watershed physical characteristics, including soil profiles and topographical features, as emphasized by (Devia et al. 2015). This detailed characterization is crucial for accurately capturing the complex interactions between infiltration, soil moisture dynamics, and surface runoff in varied catchment conditions.

Notable physical models include MIKE-SHE, KINEROS, VIC, and PRMS (Abbott et al. 1986; Todini 1988). These modeling approaches are grounded in different mathematical and physical laws (Sitterson et al. 2018). The MIKE-SHE model, known for its comprehensive approach, integrates LULC by determining evapotranspiration, runoff coefficients, and infiltration rates, allowing for a spatial representation of different land covers (Refsgaard and Storm 1995). KINEROS, an event-oriented model, emphasizes the role of LULC in overland flow generation and infiltration, representing different land covers as distinct polygons or grid cells (Goodrich et al. 2012). The Variable Infiltration Capacity (VIC) model, designed for macro-scale hydrologic simulations, uses LULC to define parameters like albedo, roughness length, and root zone depths, adopting a grid-based approach where each cell can encompass multiple LULC types (Liang et al. 1994). The soil in the model is structured into three layers: the top layer facilitates rapid soil evaporation, the middle layer captures the soil's dynamic response to rainfall, and the bottom layer depicts soil moisture behavior (Devia et al. 2015). Lastly, the Precipitation-Runoff Modeling System (PRMS) offers a deterministic, distributed-parameter perspective. LULC influences processes from snow accumulation to runoff generation, utilizing a modular approach to represent varied land covers (Leavesley et al. 1983).

2.3.3 | Challenges

Despite advances in remote sensing for land surface monitoring, models often overlook the temporal dynamics of LULC, assuming static land use for extended periods. The generation of runoff from rainfall events is a spatially and temporally complex process profoundly shaped by the terrain's topography and LULC. Modeling the effects of slope remains an unresolved issue, especially when vegetated surfaces are at play (Singh 2017).

Many models are formulated for horizontal land surfaces (i.e., SWAT does not adjust curve numbers for slope by default). Moreover, the known morphological evolution of many catchments is frequently sidelined, either for simplification or due to data constraints (Ghomash et al. 2019). The Digital Elevation Model (DEM) quality, crucial for many applications, is influenced by various factors, including land cover and terrain slope (Zhao et al. 2010; Sulis et al. 2011; Polidori and El Hage 2020).

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Landscape connectivity, which denotes how landscapes facilitate or hinder movement across them, plays a pivotal role in runoff production (Zhao and Huang 2022). It is vital to grasp as it affects hydrological outcomes, such as heightened flood risks or diminished groundwater recharge. With impervious structures like roads and buildings, urbanization disrupts natural water flow, amplifying surface runoff. In contrast, natural terrains with dense vegetation promote infiltration, curbing direct runoff. The complexity of various terrains and the nuanced configurations of landforms at different scales pose additional difficulties in creating accurate hydrological models (Gao et al. 2018; Galin et al. 2019). The role of vegetation litter in influencing surface runoff and soil erosion remains an enigma (Li et al. 2014).

Further, changes such as deforestation, urbanization, or agricultural expansion can amplify soil erosion (Nampak et al. 2018; Hu et al. 2021). Removing vegetation exposes soil to direct rainfall impact and accelerates surface runoff, which can dislodge soil particles (Zuazo and Pleguezuelo 2009; Zhao et al. 2022). As erosion progresses, nutrient-rich topsoil is lost, potentially exposing less permeable layers like clay or bedrock (Zuazo and Pleguezuelo 2009). This further complicates modeling as it can alter infiltration rates and increase runoff (Mohr et al. 2013; Wu et al. 2023).

Another critical yet often overlooked challenge is the phenomenon of soil hydrophobicity. The water-repellent nature of certain soils, often due to organic compounds from plant residues or microbial activity, can drastically reduce water infiltration rates. This, in turn, leads to increased surface runoff, posing challenges for accurate modeling (Orfánus et al. 2021).

Generalizing hydrological processes as an empirical parameter may not be representative of actual hydrological behaviors and dynamics (Sun et al. 2017). Models, such as the SCS curve number, have been critiqued for their lack of physical justification (Beven 2011). Numerical models such as Regression, Artificial Neural Network (ANN), Fuzzy, and Genetic Algorithms (GA) are data-driven models incapable of modeling spatially explicit relationships of hydrological responses concerning physical characteristics of the basin, such as the influence of change in vegetation on various hydrological components.

2.4 | Subsurface Processes

2.4.1 | Physical Mechanisms

One of the hydrological processes that can be affected by LULC change is groundwater recharge. A recharge area can be defined as that portion of the drainage basin in which the net saturated flow of groundwater is directed away from the water table (Freeze and Cherry 1979). Recharge can be categorized into several types, like natural groundwater recharge, artificial groundwater recharge, induced recharge, inter-aquifer recharge, and incidental recharge. Among the factors affecting recharge (climate, LULC, soil type, topography, and landforms), LULC has been found to be one of the key parameters in estimating recharge (Allison et al. 1990, 1994; Sandvig and Phillips 2006; Batelaan and de Smedt 2007; Crosbie et al. 2010; Kim and Jackson 2012; Li, Si, et al. 2018). The nature of the LULC, for example, urban, agricultural, or forested area, can influence the volume, quality, and type of this recharge. Seasonal LULC variations affect recharge due to the diverse recharge potential in different seasons (Siddik et al. 2022).

Urbanization often replaces permeable surfaces with impermeable ones like asphalt, reducing natural recharge (Abdelaziz et al. 2020). Conversely, human activities and infrastructure, like building artificial surface water storage structures such as dams, ponds, lakes, and canals, can enhance groundwater recharge by allowing water to infiltrate the soil below and enter the groundwater storage (Gale et al. 2002). Induced recharge can also occur when human activities, such as heavy groundwater extraction, cause surface water to seep into aquifers (Ayotte et al. 2011).

Artificial or managed aquifer recharge can augment groundwater levels (Yadav et al. 2012; Zhang, Xu, et al. 2020; Raja Shekar and Mathew 2023). Plantations may also help enhance groundwater recharge as planted landscapes are regularly rotated and monitored (Kristanto et al. 2022). Moreover, water extraction in plantations tends to be less extensive than in dense forests, consequently significantly reducing evapotranspiration (Shi et al. 2012; Geng et al. 2022). Thus, if dense forests turn into fragmented landscapes, groundwater recharge can potentially increase (Scanlon et al. 2007). In contrast, expanding impervious surfaces mean less land where recharge may occur. On the other hand, fewer trees in urban areas abstract a lot of water and lose it through evapotranspiration (Hornbeck et al. 1993).

In some places, water moves upward due to capillary forces, influencing soil moisture, while in areas with shallow water tables, phreatophytes might directly tap into this subsurface reservoir, extracting and releasing it via evapotranspiration (Batelaan et al. 2003; Wang et al. 2023). Incidental recharge results unintentionally from human activities like irrigation or canal seepage (Asano and Cotruvo 2004; Wakode et al. 2018). Agricultural practices offer a dichotomy. While irrigation can enhance groundwater recharge (Zhang et al. 2023), the expansive agricultural areas often extract groundwater at rates surpassing the recharge. Moreover, these practices can lead to soil issues such as salinization, which impede natural recharge (Kulmatov et al. 2020).

In vegetated systems, root characteristics such as form, diameter, depth, distribution, and developmental state significantly influence their function (Collins and Bras 2007; McCormack et al. 2015; Coleman and Aubrey 2018). Consequently, recharge rates demonstrate variability primarily attributed to the diverse root distributions characteristic of various topographical features, including tablelands, upslopes, midslopes, and downslopes (Collins and Bras 2007; Li, Si, et al. 2018).

Root depth and distribution are key to how plants manage soil moisture, particularly in water-limited ecosystems (Collins and Bras 2007). Deep-rooted vegetation can significantly affect groundwater recharge, especially in areas with shallow water tables (Barua et al. 2021). Due to evolutionary adaptation, for example, osmotic adjustment or hydraulic redistribution, arid and semi-arid native vegetation is capable of making

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better use of soil moisture and accessing deeper soil water reserves as compared to crops (Scanlon et al. 2005; Chen and Jiang 2010; Sardans and Peñuelas 2014). Water limitations in arid and semi-arid regions lead to a general trend where evapotranspiration rates among different vegetation types tend to be more comparable than in more water-abundant environments (Owuor et al. 2016). However, within this overall trend, there are variations. Forests and shrubs, due to their deeper root systems and physiological adaptations, are typically more efficient at transpiring water than grasslands or crops, resulting in lower recharge rates (Scanlon et al. 2002, 2005; Li, Si, et al. 2018). Consequently, non-vegetated areas, or those with less efficient water use, may experience higher recharge rates (Fitts 2013), but this, too, is subject to the influence of other factors mentioned earlier.

2.4.2 | Modeling Approaches

Modeling subsurface hydrological processes focuses on water movement through subsurface areas, including the unsaturated (vadose) zone and saturated groundwater flow. Empirical models of recharge employ advanced techniques such as geostatistics and Markov chain models to represent the variability of subsurface attributes (Yu and Lin 2015; Maples et al. 2020). Models like Kostiakov and Horton, which are either semi-empirical or empirical, are typically derived from field or lab data and are often represented as straightforward equations (Mishra et al. 2003; Ma et al. 2010). Moreover, machine learning models can draw connections between groundwater levels and associated rainfall amounts based on dominant patterns (Yifru et al. 2021).

Conceptual groundwater models typically utilize interconnected storage components or tanks, which are intrinsically tied to a water budget (Yifru et al. 2021). These tanks symbolize distinct storage zones in the soil, such as surface and root zone or unsaturated and saturated zone (Liu et al. 2017). Each zone has unique storage dynamics crucial to the soil water budget. These models assume one-dimensional vertical water flow through unsaturated zones, considering single or multiple noninteracting soil columns (Healy 2010). Transfer functions transport drainage from the bottom of the root zone to the groundwater. Conceptual models may explicitly account for plant growth, water uptake from soil by plant roots, and the release of that water to the atmosphere through leaf stomata (Fischer et al. 2008; Healy 2010).

In contrast, physically based models offer a more comprehensive view of subsurface processes. Notably, the Richards equation and the Green–Ampt model are among the most prevalent physically based models (Ma et al. 2010). The Green–Ampt model is primarily designed to describe soil infiltration (Green and Ampt 1911; Chen et al. 2015). While the Green–Ampt model does not inherently incorporate LULC factors, it can be integrated with other models or systems considering LULC changes.

Other groups of physical models rely on Darcy's law for the saturated domain, and the Richards equation for the unsaturated conditions, which describes the vertical movement of water in response to capillary and gravitational forces. LULC and root characteristics can be integrated into models based on the Richards equation by specifying distinct zones within the soil profile where hydraulic properties like conductivity and water retention curves are altered. The hydraulic conductivity itself may also be modified in root-affected zones to account for changes in soil structure (Kutílek 2004). Additionally, plant water uptake can be modeled as a sink term in the equation, with the rate of uptake influenced by both root density and soil moisture conditions (Yadav and Mathur 2008). Finally, spatial variability in vegetation types and root characteristics can be integrated using LULC data, thus making the Richards equation-based model more adept at capturing the complex interactions between root systems and soil moisture dynamics.

2.4.3 | Challenges

The relationship between LULC and groundwater recharge is often non-linear and influenced by various interacting factors like soil properties, climate, and human activities, making it challenging to capture these complex interactions accurately. Estimating infiltration at different spatial scales grows more intricate due to the natural variability in soil hydraulic properties and the range of rainfall patterns (Morbidelli et al. 2012). Adding LULC to groundwater and recharge models introduces an additional layer of complexity to this challenge, requiring an interdisciplinary, context-sensitive approach. However, by incorporating dynamic LULC data, groundwater models can become more reliable, versatile, and applicable for various temporal and spatial scales.

The integration of LULC data into subsurface process modeling is critically hampered by the limited availability of high-resolution LULC data, especially detailed information on root systems and soil interactions (Sun and Li 2023). This deficiency is significant since vegetation, a key element of LULC, substantially affects subsurface moisture dynamics. Specifically, plant roots can extract water from deeper soil layers, thereby influencing the moisture balance. These interactions are complex and vary widely across different spatial and temporal scales (Beven et al. 2015; Xie et al. 2022). For example, root structures undergo seasonal variations, and external climatic factors further complicate these changes.

Adding to the complexity is the dynamic nature of LULC, which can change over time due to factors like urbanization, agriculture, and deforestation. Rapid or extensive LULC changes can profoundly alter subsurface water pathways, storage, and availability (Foley et al. 2005; Owuor et al. 2016; Xie et al. 2022). This requires models to incorporate time-series data, a resource that is not always readily available. However, this inclusion also adds new parameters that require additional calibration, validation, and uncertainty assessment, especially in areas where observational data are sparse. Further complicating is the mismatch in spatial and temporal scales between groundwater processes and changes. This scale mismatch can result in misleading or inaccurate model outcomes if not adequately addressed (Xie et al. 2022).

Some LULCs, like urban areas, pose challenges in balancing water due to altering natural drainage patterns and widely

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distributed new abstraction points in the urbanized region (Wakode et al. 2018). Calculating overall recharge can be achieved by combining separate estimates of natural recharge from precipitation with recharge from water supply and wastewater systems (Lerner 2002; Wakode et al. 2018). However, the challenge here is unpermitted water extraction. These unlicensed groundwater abstractions introduce unknown variables, leading to inaccuracies and uncertainties in model predictions.

Moreover, specific LULC alterations, such as certain agricultural practices or the deployment of heavy machinery, can induce soil compaction, thereby affecting its porosity and permeability and, consequently, the subsurface water movement. Legacy effects from past LULC changes, like deforestation or wetland drainage, can persistently influence subsurface water dynamics (Lamichhane and Shakya 2019, 2020; Guo et al. 2020). The repercussions of historical land management decisions can reverberate, affecting subsurface hydrological processes for extended periods. Given these complexities, relying exclusively on empirical equations might fall short of accurately capturing the multifaceted impacts of LULC variability. While climate change and land cover shifts can significantly impact shallow groundwater systems, how future recharge will influence these systems is unclear (Mussa et al. 2020). Changes, particularly in built-up areas, have been a focal point in many studies, but alterations in agricultural practices often go unnoticed. In regions with intensive agriculture, a shift in crop type can significantly affect water recharge. It is essential to account for both transient (like vegetation shifts) and permanent (like urban expansion) changes (Adhikari et al. 2022).

While fundamental to vadose zone hydrology, the Richards equation has been critiqued for its inability to capture the intricacies and heterogeneities of soil water flow, prompting calls for a re-evaluation and the development of more comprehensive measurement techniques. It assumes a soil continuum in basic form, potentially missing small-scale soil variations. The equation excludes tortuous water pathways, air entrapment, and plant root effects. Its reliance on the Darcy-based relationship for unsaturated flow also overlooks specific kinematic effects and dynamics between liquid and gas phases (Beven and Germann 1982, 2013; Beven 2011, 2018). It also may not capture preferential flow paths in soils with features like cracks (Beven 1989; Binley, Beven, et al. 1989; Binley, Elgy, et al. 1989). Beven (2019a) states that the Richards equation should be reconsidered, rooted in an experiment that ignored preferential flows. This stance is backed by both experimental evidence and inherent physics, especially considering soil property heterogeneity (Beven and Germann 1982, 2013; Beven 2011, 2018). Flows, in reality, can be localized, variable, and have spatial–temporal complexities (Freer et al. 1997; Jencso et al. 2009; McGuire and McDonnell 2010; Klaus and Jackson 2018). Yet, we lack effective measurement techniques for studying such flows. Instead of focusing on intricate details, there is a suggestion to develop measurement techniques for larger scales that encapsulate these details (Beven and Germann 1982, 2013; Beven 2011, 2018).

3 | Discussion

3.1 | LULC Data

Space agencies are deploying satellites equipped with advanced sensors for LULC mapping, utilizing Earth Observation (EO) data (Petrişor et al. 2010; Diaz-Pacheco and Gutiérrez 2014; Pandey et al. 2021). Satellite imagery is characterized by various attributes, including spatial, spectral, and temporal resolutions, which determine its suitability for diverse applications. Higher spatial resolution enhances mapping accuracy by providing detailed images, while increased spectral resolution assists in distinguishing different features (Pandey et al. 2021). Temporal resolution is equally important for detecting changes and understanding dynamics over time. For instance, seasonal resolution is vital for monitoring agricultural cycles, while annual resolution helps in urban and forest cover studies. Increasing remote sensing data has led to the emergence of new classification systems for evaluating LULC and spatial changes. Understanding the input dimensions, types of remotely sensed data, and the correct implementation of classifiers in LULC mapping is essential. This understanding enables one to effectively assess the advantages and limitations of various approaches and their combinations as applied in the study (Pandey et al. 2021). Detailed information regarding Earth observation data sources and classifiers can be found in the study by Pandey et al. (2021).

Various sources of LULC data exist worldwide (Pérez-Hoyos et al. 2012; Szarek-Iwaniuk 2021). Among the most prominent ones are the Corine Land Cover (CLC) system, developed by the European Environment Agency for Europe's environmental monitoring (Büttner 2014); the United States Geological Survey (USGS) Land Use and Land Cover Classification System (Witmer 1978), used in the United States for standardized land classification; and the FAO Land Cover Classification System (LCCS) (Herold and Schmullius 2004), offering a flexible framework adaptable to various regions and scales. Additionally, the Modified UNESCO Classification (Becker et al. 1998), the International Geosphere-Biosphere Programme (IGBP) Land Cover Classification (Loveland et al. 2000), the Global Land Cover (GLC) project (Bartholome and Belward 2005), and the MODIS Land Cover Type play crucial roles in supporting global environmental research and providing comprehensive land cover data (Friedl et al. 2002).

While LULC products are diverse and accessible, they have limitations like limited coverage (both temporal and spatial), data validity, detail level, and inconsistent nomenclatures. LULC databases are updated inconsistently and may not meet current research demands (Pandey et al. 2021; Szarek-Iwaniuk 2021). Moreover, many landscapes and sensors have spawned varied classification techniques, complicating comparative analyses. Further, many LULC maps are overly generalized, lacking the detail needed for specific research (Pandey et al. 2021; Szarek-Iwaniuk 2021). This simplified categorization disregards the inherent variations within the same category, which can significantly influence vegetation's water storage and transportation mechanisms. These factors

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impose constraints on the efficacy of hydrological models, potentially causing them to succeed in specific case studies while yielding suboptimal results in others. To overcome this limitation, data fusion techniques can be employed to integrate information from multiple sources with varying spatial and temporal resolutions. By combining data from different satellite platforms—such as optical sensors like Landsat and Sentinel-2 with higher temporal frequency sensors like MODIS or radar systems not affected by cloud cover—it's possible to create more temporally continuous and spatially detailed LULC datasets.

A crucial improvement lies in augmenting the current LULC data with additional information to enhance surface characterization. Without that, sparsely vegetated boreal evergreen needleleaf forests could be functionally interchangeable with a coastal temperate evergreen needleleaf forest as they may fall into the same LULC category. Even if labeled similarly, LULC can exhibit vastly different hydrological behaviors depending on the context, such as variations in the understorey and litter layers and temporal factors like seasonal changes. This generic categorization often results in models exhibiting sensitivity based on parameter settings and temporal scales (Huang et al. 2013; Jin et al. 2019).

3.2 | Integration of Dynamic LULC

Despite advancements in vegetation understanding and remote sensing, many models inadequately incorporate LULC dynamics (Sun et al. 2017). Even with data allowing for LULC temporal variations, many models persistently use static LULC. However, it is essential to integrate continuous LULC changes for accurate LULC-climate-hydrology interactions in a rapidly changing world (Castillo et al. 2014). Accurate and consistent mapping of change dynamics requires attention to aspects from spatial to spectral and temporal dimensions. In large-scale hydrological studies, pre-modeling data analysis is crucial to understanding the dynamics and inconsistencies between LULC datasets (Kauffeldt et al. 2013). Analyzing different datasets independently aids in forming robust conclusions on choosing the best model to achieve the desired outcome (Juston et al. 2013; Magnusson et al. 2015).

After ensuring that LULC classes are consistently defined across different time periods to avoid discrepancies in model inputs, the next step is to relate these LULC classes to hydrological parameters—such as roughness coefficients, infiltration rates, and evapotranspiration rates—that the model uses to simulate hydrological processes (Zhang, Wang, et al. 2020; Demissie 2022; Chen, Chi, et al. 2023). Incorporating routines within the model to update these parameters as LULC changes over time can be achieved by creating time-series inputs that the model reads at specified intervals. Various coupling methods can be employed to effectively integrate dynamic LULC data with hydrological models. In unidirectional coupling (Liu, Zhang, et al. 2024; Jiang et al. 2021), LULC data are used as inputs in the hydrological model without feedback, allowing the model to consider land cover changes passively. Introducing a feedback loop enables changes in the hydrological model to influence the LULC data, such as how changes in water availability might impact vegetation growth. Dynamic bidirectional coupling updates the connection between the LULC and hydrological models in real time, enabling a more flexible and accurate simulation of hydrological processes as they happen (Liu, Wang, et al. 2024). Machine learning techniques enhance this coupling by identifying complex, non-linear relationships between LULC changes and hydrological responses.

3.3 | Integration of Remote Sensing Data

Coupling dynamic LULC data with hydrological models significantly enhances the temporal resolution and accuracy of hydrological predictions, especially in rapidly urbanizing or deforested areas (van Meerveld and Seibert 2024). Traditional models often rely on static LULC parameters, overlooking the unique and evolving hydrological characteristics of each LULC type (Zhang et al. 2019; Zhang, Wang, et al. 2020). By integrating real-time satellite-derived LULC data, models can dynamically update parameters to reflect immediate landscape changes accurately. For instance, advancements in satellite remote sensing have ushered in a new era for monitoring evapotranspiration, particularly on broader scales with heightened spatio-temporal precision (McShane et al. 2017). Beyond mere data collection, remote sensing has proven instrumental in discerning snow cover nestled within forest canopies, offering a more profound understanding of snow interception dynamics (Allen et al. 2020). Other examples are the C- and K-factor model parameters in the Universal Soil Loss Equation (USLE), which can be recalculated dynamically using NDVI time series data and changing soil conditions derived from remote sensing. Additionally, parameters like curve number, Manning's roughness coefficient, Fractional Vegetation Cover (FVC) and LAI can be linked to specific LULC classes and updated as land use changes, enhancing the representation of surface runoff, infiltration, and evapotranspiration processes. Incorporating remote sensing vegetation parameters can significantly improve the simulation accuracy of key processes like evapotranspiration, runoff, and soil moisture (Chen, Fu, et al. 2023). By dynamically updating LAI within the model, vegetation-dependent processes such as water uptake and surface shading are better accounted for, leading to enhanced predictions of hydrological fluxes and a more accurate representation of seasonal vegetation growth cycles (Zhang et al. 2019; Chen, Fu, et al. 2023). Similarly, FVC can be calculated based on NDVI and plays a critical role in determining the partitioning of evapotranspiration into vegetation transpiration and bare soil evaporation (Ji et al. 2024). The dynamic adjustment of FVC allows models to capture temporal shifts in ET partitioning and soil moisture, especially in regions undergoing active vegetation changes, such as agricultural zones or recovering forests (Ji et al. 2024). Moreover, FVC alongside LAI improves the simulation of soil moisture dynamics, as it provides a clearer depiction of root-zone water uptake, which directly influences groundwater recharge and runoff generation processes (Yang et al. 2016; Zhang et al. 2018). This dynamic parameterization allows for the exploration of temporal variability across LULC classes, resulting in predictions that are more responsive to landscape changes.

Remote sensing vegetation data can also be used for the calibration and evaluation of hydrological models. For instance,

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Chen, Fu, et al. (2023) demonstrated that integrating remote sensing-detected phenology into models like SWAT enhances the simulation of LAI, reducing uncertainties in vegetation growth and water balance components at watershed scales. Data assimilation techniques that incorporate real-time remote sensing data enable dynamic updates of hydrological model states, enhancing simulation accuracy. For example, Kalu et al. (2022) demonstrated that coupling deep learning with data assimilation improves the modeling of TWS dynamics, offering a more robust approach to handling complex climate and hydrological interactions. Additionally, Zhang et al. (2023) proposed a hybrid model combining physically based and data-driven models for real-time groundwater level forecasting. While physically based models capture intricate hydrological interactions, they require extensive data and can be computationally intensive. In contrast, data-driven models, particularly deep learning and machine learning approaches, efficiently capture nonlinear relationships without explicitly defining each hydrological process (Shen 2018). Hybrid models represent a promising approach, balancing the physical consistency of traditional models with the flexibility of data-driven techniques. By assimilating remote sensing data into these hybrid models, we can dynamically update model states based on real-time inputs, thereby reducing uncertainties in hydrological forecasts and improving predictions in non-stationary and ungauged regions. While these technological strides provide significant insights, ground observation data continue to be pivotal for the calibration of such imagery, solidifying the trustworthiness of subsequent forecasts (Chia et al. 2020). This improvement in satellite innovation and ground observations is a promising pathway to refine and enhance hydrological models.

3.4 | Selecting the Appropriate Model

While modeling every physical process is not viable, it is essential to understand desired outputs and choose an optimal model structure based on data and budget constraints. Process simplification results from limited knowledge and data, imprecise measurements, and the involvement of multiple scales and interactive processes. These simplifications introduce uncertainty and make it an intrinsic property of any model (Moges et al. 2020). More intricate models prove beneficial for exploring natural hydrological principles and mirroring real-world processes with higher fidelity (Zhao et al. 2013).

However, model performance does not always indicate increased accuracy; outcomes can significantly vary based on factors like the specific hydrological variable (e.g., runoff vs. soil moisture), hydrological conditions (floods vs. droughts), and temporal scale (Orth et al. 2015). Complex models may suffer from over-parametrization but miss relevant processes if they are too simple (Orth et al. 2015). For example, Lopez et al. (2020) demonstrated that setting up an energy-balance model with a simplified snowpack structure can provide nearly identical performance as a much more complex snow-physics model.

Model complexity is vital while evaluating the ability of the models to simulate the desired LULC and climate change scenarios. However, when more detailed results are needed, a fully distributed and/or physically based approach may be required, and it may be necessary to collect detailed data to apply the model (Bormann et al. 2009; Elfert and Bormann 2010). Besides spatial discretization, the model's temporal resolution, which may be restricted to specific intervals (e.g., sub-daily, daily, monthly), significantly impacts data availability and model complexity (Chien et al. 2013).

Differences in simulation quality among models arise from uncertainties in input data, calibration approaches, parameterization, and model structures (Cornelissen et al. 2013; Dwarakish and Ganasri 2015). Therefore, it is essential to be aware of the assumptions, trade-offs, and sources of error inherent to each downscaled data set and understand how those issues might apply to the particular question and location (Singh 2017). Researchers must apply models judiciously, re-evaluating processes like the Buckingham–Richards, as suggested by Beven (2019a). Beven further underscores that uncertainties are a fundamental characteristic of hydrological data (Beven 2019a, 2019b). These uncertainties are frequently treated as aleatory, a classification largely driven by the ease of applying statistical techniques for analysis.

The broader modeling landscape, especially related to LULC, is undergoing significant shifts. For instance, the widespread online availability of hydrological data facilitates expansive and efficient simulations with platforms like Google Earth Engine. This accessibility, combined with diverse modeling approaches, not only refines our comprehension of hydrological processes but also elucidates the nuanced interplay between LULC changes and climate shifts (Cornelissen et al. 2013).

Moreover, while it is challenging to account for all processes associated with LULC, their effects are evident in the data. Advanced analytical techniques, enhanced by data-mining innovations, have the potential to offer more universally relevant solutions and introduce alternative process perspectives. Given the limitations of empirical, physics-based, and conceptual models, there is a growing interest in advanced data-driven models, including machine learning (ML) and deep learning (DL) (Jehanzaib et al. 2022).

3.5 | Enhancing Model Mechanisms

While remote sensing offers qualitative insights into landscape patterns, it carries epistemic uncertainties (e.g., rainfall and soil moisture estimation). Complexities arise in capturing soil variations, antecedent moisture, infiltration, topography, and rainfall distribution. Further, dealing with human-made infrastructures and vegetation is still associated with many uncertainties. LULC significantly impacts hydrological processes in models with natural LULC changes typically evolving gradually because of ecological processes. In contrast, human activities like deforestation, agricultural expansion, and urbanization cause rapid LULC alterations. Significant natural shifts, such as those from bushfires or floods, can arise due to extreme climate events.

These rapid changes, whether from human activity like dam construction or natural events like bushfires, can profoundly impact landscapes and the water balance of an area (Singh 2017). The rapid and significant changes in vegetation

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and climate violate the prevailing assumption of a long-term steady water balance and stationary assumption in hydrological models (Milly et al. 2008). This presumption, which does not account for abrupt or substantial environmental shifts, introduces considerable uncertainty in the water balance, particularly in regions experiencing extensive LULC transformations (Devia et al. 2015; Liu et al. 2019).

Understanding the hydrological response to rapid LULC changes is challenging (Dwarakish and Ganasri 2015). In many studies, large and frequent inconsistencies between climate datasets and observed discharge showed clear spatial patterns, possibly caused by anthropogenic influences (e.g., inter-basin transfers, irrigation and reservoirs) (Devia et al. 2015; Liu et al. 2019). Infrastructure and urbanization can alter surface connectivity, potentially increasing runoff and flooding risks (Wang et al. 2022).

While some changes may seem minor or localized, their ripple effects on the hydrological processes can be substantial. For instance, the escalation of erosion due to deforestation in sloped areas results in an accumulation of sediments in aquatic systems (Brandolini et al. 2018; Kumar et al. 2022). This process alters flow regimes and storage capacities and degrades water quality, presenting additional challenges in hydrological modeling and water resource management.

One of the challenges in incorporating dynamic LULC is that LULC changes impact hydrological components across varied timeframes. For example, deforestation instantly eliminates interception, but its effects on groundwater recharge manifest later. Enhancing the quality and quantity of hydrological data on both spatial and temporal scales is pivotal to capturing these nuances.

The intricacies of evapotranspiration, especially in forests, remain an area for enhancement in hydrological models. Transpiration is dominant, yet interception evaporation, especially in forests, can sometimes surpass it (Wang-Erlandsson et al. 2014; Wei et al. 2017). Modern modeling often overlooks forests' ability to store heat and vapor across multiple layers, from the air column to the soil (Coenders-Gerrits et al. 2020). Studies should delve deeper into the canopy interactions and understand water, vapor, and heat dynamics to better grasp forest evaporation. Future efforts should prioritize understanding the interactions among leaf growth, abscission, leaf litter dynamics, and evaporation partitioning (Mert 2021). Additionally, understanding the processes within and beneath the canopy and how evaporation splits between interception, transpiration, and soil evaporation is crucial (Dubbert et al. 2013; van den Hoof et al. 2013). Incorporating stable water isotopes can aid in comprehensive modeling from vegetation top to forest floor.

LULC influences the hydrological processes through manifold (Afonso de Oliveira Serrão et al. 2022; Siddik et al. 2022). Land cover dictates soil erosion, with forests acting as protective barriers. Urbanization complicates channel flows, while deforestation affects snowpack dynamics. LULC also modulates water quality, with agricultural lands potentially introducing nutrients and urban zones contributing to pollutants (Roy et al. 2022; Gu and Li 2024). Drained or altered wetlands disrupt regional water balances and flood mitigation. Soil moisture, key to the critical zone and, hence, the hydrology, is directly linked to LULC.

Moreover, critical aspects like hydraulic conductivity of aquifers and roughness of the land surface are inadequately addressed spatially. While these challenges are recognized, hydrological models might miss some nuances. For instance, the temporal dynamics of soil hydrophobicity, the micro-scale impacts of LULC on soil structure, or the feedback loops between erosion, landscape connectivity, and local hydrology might not be fully captured (Blume et al. 2009; Stephens et al. 2021; Vereecken et al. 2022). Addressing these gaps requires a holistic approach, integrating interdisciplinary knowledge and continuously updating models based on new findings.

The modeling community needs to access web services from data providers and improve the spatial resolution of data to account for the great variability in climate, topography, and land cover (Sitterson et al. 2018). The availability of the model and data to the public increases their usefulness, reduces duplication of efforts, and saves time and money (Sitterson et al. 2018). Also, novel remote sensing products such as vegetation optical depth may provide additional useful information on global scales of canopy structure (Rodríguez-Fernández et al. 2018). Overall, capturing LULC nuances in hydrological models is essential for accurate predictions, urging continued improvements in understanding these interplays. Future studies should examine potential shifts in the LULC pattern, considering both biophysical and socio-economic factors in a catchment with diverse vegetation (Dwarakish and Ganasri 2015).

3.6 | Model Evaluation and Calibration

In the calibration process of hydrological models, the direct alteration of LULC data is not typical. Instead, the focus is usually on adjusting model parameters rather than improving the representation of process dynamics (Rajat and Athira 2021). However, aspects connected to LULC are often fine-tuned or more deeply scrutinized to achieve optimal results (Sai Veena et al. 2019). That may improve the reproduction of discharge without change, albeit not always for the right reasons (Beven 2019a). For instance, the available LULC data might be reclassified or adjusted better to fit the model's specific requirements or resolution. Obtaining better results for the wrong reasons may lead to misleading interpretations, especially when simulating the impacts of LULC change (Merz et al. 2011; Peel and Blöschl 2011; Hollaway et al. 2018).

The poor process dynamics in the model reduce its predictive power (Rajat and Athira 2021). Several hydrological parameters, such as curve numbers or Manning's "n" values, are intrinsically tied to specific LULC types. These parameters can be refined during calibration to improve model fit.

Hydrological systems are nonlinear and complex, and we have few techniques for studying patterns of processes at the catchment scale. In many instances, models have shown different sets of calibration parameters for different periods (Merz et al. 2011). Therefore, identifying and understanding the time stability of catchment model parameters and how they may change

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with time need to be understood better (Wagener et al. 2010). Traditionally, hydrological models have assumed stationary conditions (the model parameters do not change with time). Therefore, accounting for temporal changes in the model parameters is not straightforward (Peel and Blöschl 2011). This is mainly because it is not easy to find a unique relationship between the changing parameters and complex correlations among the parameters and with various catchment characteristics (Wagener 2007; Merz et al. 2011). All these assumptions can produce significant errors in the simulation process if climate and/or catchment conditions change much over time (Merz et al. 2011). These errors potentially extend with increasing the scale and time lag between the calibration and simulation periods.

Incorporating dynamic LULC into hydrological models substantially influences the calibration process, offering both challenges and advantages. Dynamic LULC introduces the need for time-variable parameters, ensuring that associated hydrological parameters like runoff coefficients evolve correspondingly as land cover changes. This necessitates a calibration dataset that spans diverse temporal conditions, capturing the intricacies of changing landscapes. While this more realistic portrayal of land changes enhances the model's applicability, it complicates calibration by increasing the number of tenable parameters, potentially elevating model uncertainty. Such increased complexity demands a rigorous validation process, testing the model's proficiency across various LULC change periods.

Furthermore, the risk of overfitting emerges, where the model might echo the calibration data closely but falter with new independent datasets (Fowler et al. 2016, 2018; Dakhlaoui et al. 2017; Pool et al. 2017). As the model's sensitivity to parameters can oscillate with changing LULC, periodic sensitivity analyses become pivotal. Merging data from assorted sources, like remote sensing and land surveys, can pose challenges, emphasizing the need for consistent scale, resolution, and accuracy. Calibrating a model with point data and assuming that some simple power law will hold over the range of the data is essentially misleading (Westerberg et al. 2020). Lastly, dynamic LULC often ushers in scenario-based calibrations, enabling the exploration of potential land management or future trajectories. While undeniably intricate, including dynamic LULC ensures models stay in step with real-world land changes, granting them enhanced relevance and reliability when appropriately calibrated (Yonaba et al. 2021).

Regions with substantial LULC shifts during the calibration period necessitate acknowledging these changes for accurate modeling (Birhanu et al. 2019). For instance, Birhanu et al. (2019) demonstrated that in regions undergoing significant LULC changes, the HBV model—which was both accurately calibrated and validated—exhibited only minor and insignificant fluctuations (± 5%) in projected water balance components like discharge, evapotranspiration, soil moisture, and groundwater recharge. However, when the actual discharge data was statistically assessed, noticeable increasing trends were observed throughout the study. It shows that static data can impede successful calibration if a model does not factor in such dynamics and there are notable LULC alterations. Further nuances, like distinct management practices on similar land covers, might also be incorporated to boost calibration accuracy. While LULC data is rarely manipulated directly, its associated parameters and processes are vital in refining hydrological models.

Most hydrological models are traditionally calibrated against observed streamflow, which can be insufficient in poorly gauged regions (Huang et al. 2020). Given that hydrological signatures often arise from multiple catchment processes, it is critical to identify which signatures best represent specific processes and effectively integrate satellite-based indicators—such as vegetation indices, LAI, ET, and TWS—which have shown potential to improve model reliability (Chen, Fu, et al. 2023; Zhong et al. 2024). Qi et al. (2020) demonstrated a framework that utilizes bias-corrected remotely sensed ET to improve model calibration and predict daily and monthly runoff. However, the limited impact of adding GRACE data—often due to its coarse spatial resolution—highlights the need for further research into its applicability in runoff modeling. Future research should focus on testing GRACE and other satellite datasets across a range of catchment scales and climate zones to better understand how remote sensing data limitations interact with hydrological model resolutions.

Advances in remote sensing, such as the improved spectral resolution of the Sentinel-2 satellite, have shown promise in detecting water stress in vegetation (West et al. 2018). Developing standardized methodologies for integrating these data types across diverse hydrological models and regions could address limitations in current approaches. This would provide a foundation for applying multi-source constraints to improve predictive performance across varied climatic and geographical contexts (Pool et al. 2024). Reducing uncertainties in remote sensing data is equally important for accurate hydrological modeling. Remote sensing datasets vary in quality, and biases between different datasets can hinder reliable predictions, especially in critical variables like ET and TWS.

4 | Conclusion

This article has endeavored to present an overview of LULC-related mechanisms involved in hydrological processes, their modeling approaches, and the shortcomings of the models in representing processes.

Physically based semi-distributed and distributed models have gained prominence for their efficacy in studying the impacts of heterogeneous LULC patterns. Nonetheless, the field of hydrology continues to face substantial constraints due to limitations in observational data. The responses of hydrological processes to LULC changes are notably diverse, shaped by a multitude of factors, including geographical attributes and other physical factors such as slope and soil characteristics.

This complexity is further compounded by human interventions and the nuanced relationship between evaporative factors and vegetation canopies, posing significant challenges for the modeling community. In addressing these challenges, contemporary research leverages a combination of field data, experimental

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studies, and sophisticated modeling techniques. The advent of remote sensing has markedly enhanced our comprehension of LULC dynamics. However, fully grasping the hydrological implications of changes requires an interdisciplinary strategy that extends beyond conventional methodologies. Embracing innovative technologies and integrated modeling approaches will be instrumental in advancing watershed management practices.

Despite the progress in hydrological modeling, several challenges persist. Current models, while insightful, struggle to overcome observational constraints and often fall short in effectively utilizing remote sensing data or accurately representing hydrological complexities. Traditional empirical formulas, born in data scarcity, impede modern modeling efforts, particularly in rapidly changing landscapes. Specific processes, such as snow-related interception, evapotranspiration, and soil moisture estimation, remain inadequately represented, particularly in multi-layered environments like forest areas.

Models integrate various physical processes with certain simplifications and empirical elements, making them a hybrid of purely empirical and fully physically based models. However, balancing between them is essential as this hugely impacts all subsequent processes. For example, existing limitations in soil water extraction functions hamper the full potential of the Penman–Monteith method.

To move forward, hydrological science requires a standardized LULC classification system to facilitate uniform modeling and enhance comparison across studies. A comprehensive modeling approach is essential, incorporating extensive validation, innovative process depiction, uncertainty management, and a nuanced recognition of data constraints. Continued exploration into the responses of hydrological processes to LULC changes, coupled with refined uncertainty quantification in parameter estimates and modeling, will remain at the forefront of hydrological research.

As we progress, it becomes imperative to adopt a more critical perspective on modeling practices, acknowledging the constraints imposed by data availability, study scope, desired output, and resource allocation. A concerted effort within the hydrological community is necessary to advance these models as viable hypotheses, fostering new process representations and effectively managing predictive uncertainties. A systematic evaluation of potential error sources, encompassing input data, boundary conditions, model design, and software assumptions, is crucial for developing reliable simulation models. The journey through hydrological research is replete with uncharted territories, from pioneering observational techniques to theoretical breakthroughs and explorations of diverse ecological landscapes. The path ahead for LULC and hydrological modeling is one of embracing technological advancements and methodological rigor, ensuring our strategies are comprehensive and genuinely representative of the multifaceted realities we seek to simulate.

Author Contributions

Amirhossein Shadmehri Toosi: conceptualization (equal), investigation (lead), methodology (equal), visualization (lead), writing – original draft (lead), writing – review and editing (equal). **Okke Batelaan:** conceptualization (equal), investigation (equal), methodology (equal), project administration (lead), resources (lead), supervision (lead), writing – review and editing (equal). **Margaret Shanafield:** conceptualization (equal), investigation (equal), methodology (equal), supervision (equal), writing – review and editing (equal). **Huade Guan:** conceptualization (equal), methodology (equal), supervision (supporting), writing – review and editing (supporting).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

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